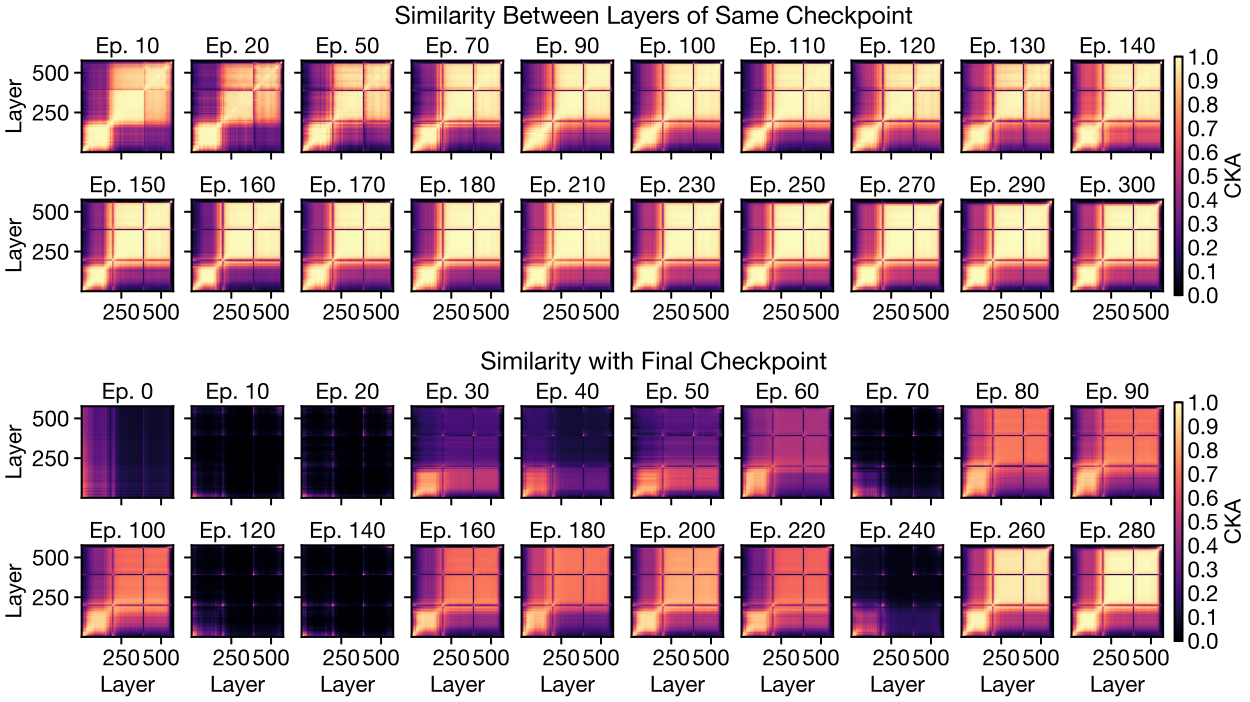

Figure F.3: Evolution of the block structure over the course of training for a ResNet-164 (1×) model. We compute the CKA between all pairs of layers within a ResNet-38 (10×) model at different stages of training, and find that the internal representations already contain a block structure at epoch 10. Comparing these different model checkpoints to the final, fully-trained model reveals that the block structure representations at different epochs are considerably dissimilar, especially during the first half of the training process.

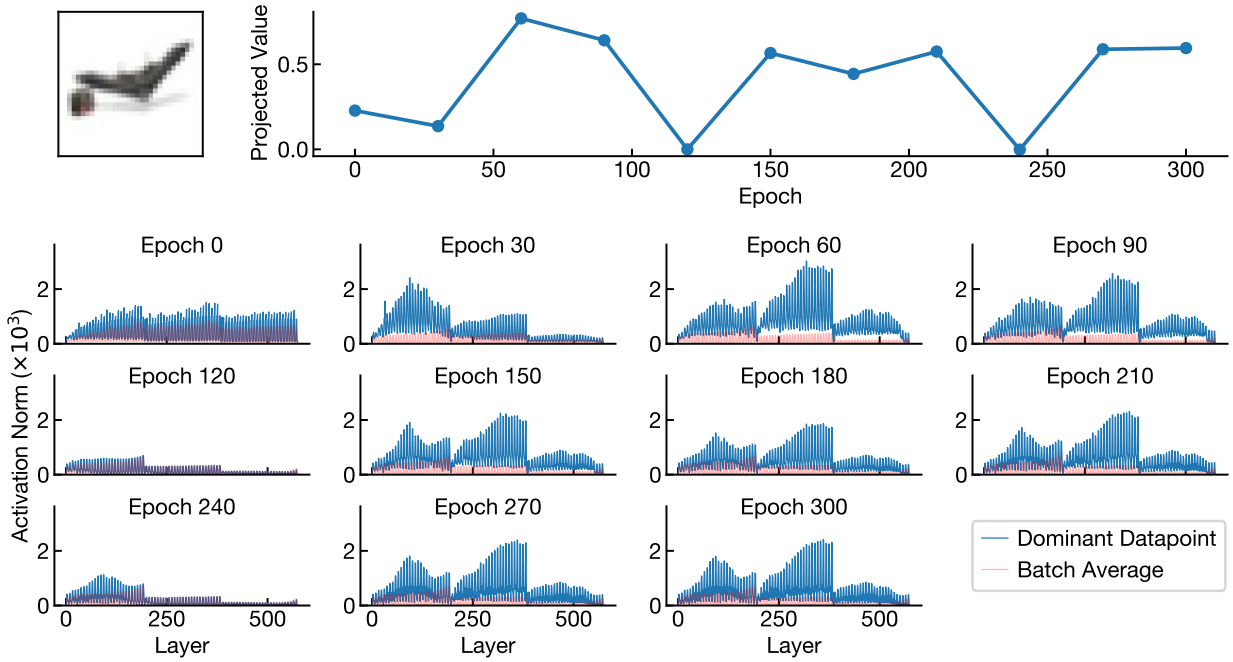

Figure F.4: Example of how the magnitudes of layer activations and the projected values onto the first principal component for a dominant image vary across different epochs, for a ResNet-164 (1×). Given a dominant datapoint for a ResNet-164 (1×) model, we track the magnitude of its projected value onto the first principal component of the block structure representations (top left), as well as its activation norms at each layer in the network (bottom set of plots), over time. Notice the correspondence between these 2 metrics, especially when their values drop at epochs 0 (initialization), 120 and 240. This is also aligned with the measurement of CKAs across different epochs (see Figure F.3, where we find that the model checkpoints at these 3 epochs are highly dissimilar from the final model in terms of the hidden representations). See Appendix Figure F.6 for the corresponding visualization of a dominant example of a wide ResNet (ResNet-38 (10×)).

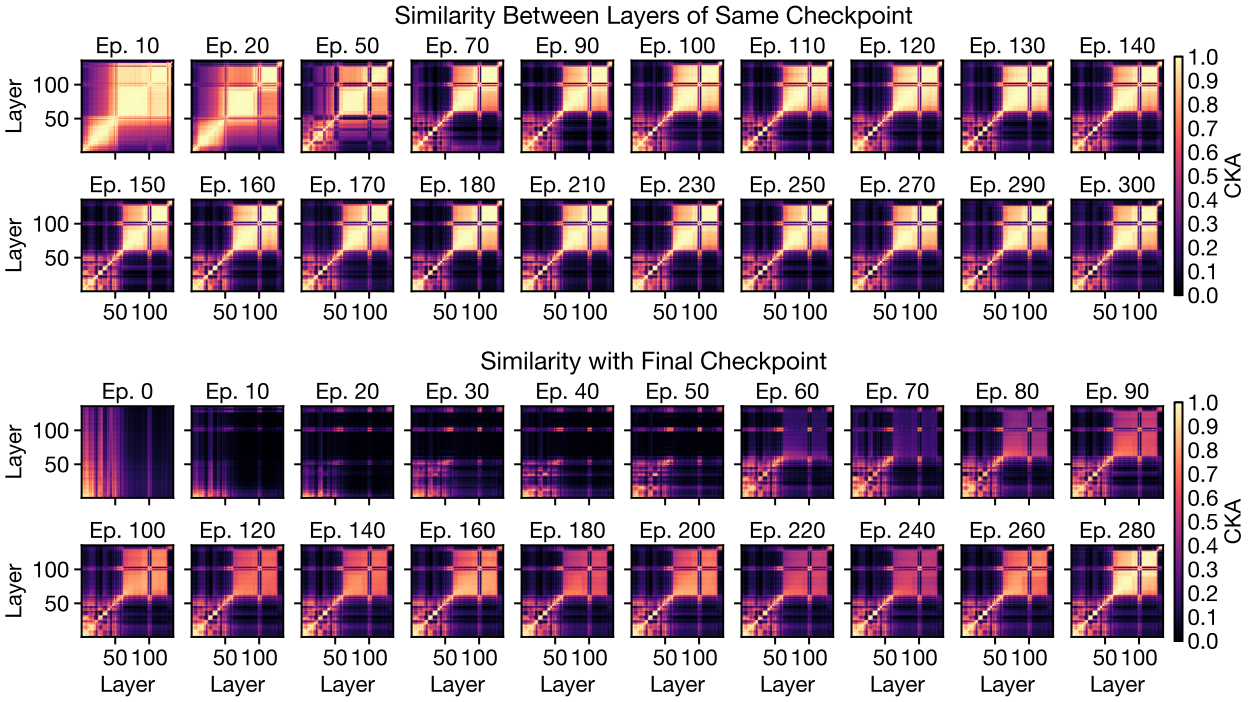

Figure F.5: Evolution of the block structure over the course of training for a ResNet-38 (10×) model. We compute the CKA between all pairs of layers within a ResNet-38 (10×) model at different stages of training, and find that the internal representations already contain a block structure at epoch 10. Comparing these different model checkpoints to the final, fully-trained model reveals that the block structure representations at different epochs are considerably dissimilar, especially during the first half of the training process. See also Figure F.3 for a similar plot for a deep ResNet (ResNet-164 (1×)).